



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Low-power Bluetooth SoC

The ATM33 series Bluetooth 5.3 family of high-performance system-on-chip (SoC) enhances Atmosic's advanced



energy harvesting and ultra-low power technology. It integrates a power optimised Arm Cortex M33F that can operate up to 64MHz and supports Arm TrustZone for robust security. The SoC series has improved energy harvesting efficiency, a wider harvesting input range, a lower cold start voltage, lithium-ion battery support, and low-power voice processing. It can perform functions such as access memory and transmit beacons even while in deep sleep. The eco-friendly SoC series supports unparalleled battery life and battery-free operation.

Atmosic
www.atmosic.com

Tactile sensor

Tactaxis is a fully integrated tactile sensor that is compact, soft, and provides a 3D force vector acting on its surface. This improves robots' hands and



grippers, making delicate operations such as fruit picking possible. It features multiple 3D magnetometer pixels and is accompanied by a magnet which is embedded into an elastomer material. This presents a soft contact interface, emulating the attributes of human skin. The arrangement offers a high sensitivity so that detection of even small amounts of force generates a response. The sensor equips robots with the sense of touch to manipulate fragile objects.

Melexis
www.melexis.com

4D lidar with high resolution

The Aeries II is a 4D lidar sensor built with automotive-grade reliability to enable the next wave of autonomy across applications in automotive,

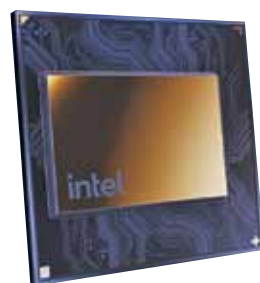


industrial, and beyond. Leveraging Frequency Modulated Continuous Wave (FMCW) technology and lidar-on-chip module design, the sensor uniquely detects the fourth dimension of instantaneous velocity for each point in addition to 3D position. Advanced 4D Perception software powers ground-breaking new features like 4D localisation and ultra-resolution, a camera-level image with up to 20 times the resolution of legacy lidar sensors. Built for reliability with automotive-grade ratings, the device ensures peak performance across a variety of road and environmental conditions.

Aeva
www.aeva.com

Chip for crypto mining

Intel Blockchain Accelerator chip is designed for crypto mining. It can offer 1000x better performance per watt



than mainstream GPUs on tasks like SHA-256 based mining. Existing ASICs consume a lot of power while mining, which now appears to be solved with the upcoming custom chip for blockchain-specific applications. The chip is designed to also solve the energy consumption problem by mining as it is a scalable and sustainable solution for replacing normal mining GPUs with custom-made blockchain accelerated chips. The Blockchain Accelerator Chip is implemented on a tiny piece of silicon, which does not affect mainstream computer processors like CPU and GPU.

Intel
www.intel.com

AI chip

The new chip's architecture delivers the performance required for next-generation applications such as AI model training, HPC, drug discovery,

